U.G.C. NET Ex
COMPUTE
(Solved with Expla

1. **Beam-penetration and shadow-mask are the two basic techniques for producing color displays with a CRT. Which of the following is not true?**
   - **I. The beam-penetration is used with random scan monitors.**
   - **II. Shadow-mask is used in raster-scan systems.**
   - **III. Beam-penetration method is better than shadow-mask method.**
   - **IV. Shadow-mask method is better than beam-penetration method.**

   (a) I and II (b) II and III
   (c) III only (d) IV only

Ans. (c) : Because, There are two ways to display an object on the screen : Raster scan and Random scan and the main method for producing color in CRT Monitors are

(1) Beam Penetration method

(2) Shadow mask method

⇒ Difference between beam penetration and shadow mask method.

	Beam Penetration	**Shadow Mask**
Where used	It is used with Random scan system to display color.	It is used with Raster scan system display color.
Color	It displays only four colors i.e Red, green, orange, yellow	It can display million of colors.
Cost	It is less expensive as compared to shadow mask	It is more expansive than others
Picture Quality	Quality of picture is not so good.	Quality of picture is so good.

2. **Line caps are used for adjusting the shape of the line ends to give them a better appearance. Various kinds of line caps used are**

   (a) Butt cap and sharp cap
   (b) Butt cap and round cap
   (c) Butt cap, sharp cap and round cap
   (d) Butt cap, round cap and projecting square cap

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Ans. (d) : Because,
Butt Cap : Chops off the stroke at the ends.
Round Cap : Extends the stroke past the ends with semicricular ends.
Projecting Cap : Extends the stroke past the ends with squared ends.

3. Given below are certain output primitives and their associated attributes. Match each primitive with its corresponding attributes :

	List–I		List–II
a.	Line	i.	Type, Size, Color
b.	Fill Area	ii.	Color, Size, Font
c.	Text	iii.	Style, Color, Pattern
d.	Marker	iv.	Type, Width, Color

Codes :

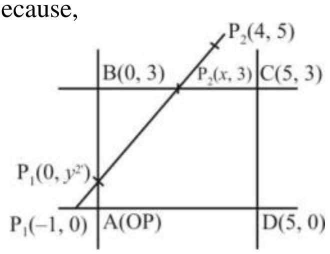
	a	b	c	d
(a)	i	ii	iii	iv
(b)	ii	i	iii	iv
(c)	iv	iii	ii	i
(d)	iii	i	iv	ii

Ans. (c) : Because,
Line : It has attribute like color, line width.
Fill Area : We can fill a region with some color with different style and patterns.
Text : Any texts can be written with different font, color etc.
Marker : The graphics attributes are the marker color, marker type, marker size. So the answer is option (d).

4. Consider a window bounded by the lines : x = 0; y = 0; x = 5 and y = 3. The line segment joining (–1, 0) and (4, 5), if clipped against this window will connect the points

(a) (0, 1) and (2, 3) (b) (0, 1) and (3, 3)
(c) (0, 1) and (4, 3) (d) (0, 1) and (3, 2)

Ans. (a) : Because,



First find the equation of line segment joining (−1, 0) and (4, 5)

$$4-0 \Rightarrow \frac{5-0}{4-(-1)}\left(x-(-1)\right)$$

$$\Rightarrow 4 = x + 1$$

Now only choice A(0, 1) & (2, 3) satisfy this, Hence the answer is (a).

5. **Which of the following color models are defined with three primary colors?**
(a) RGB and HSV color models
(b) CMY and HSV color models
(c) HSV and HLS color models
(d) RGB and CMY color models

Ans. (d) : Because,
RGB : Red, Green, blue
CMY : Cyan, magenta, yellow, black is a subset of RGB.
HSV : HUC, Saturation, Value (also known as HSB : HUC, Saturation, Brightness).
HSL : HUC, Saturation , lightness/luminance
Hence the answer is (d).

6. **In a digital transmission, the receiver clock is 0.1 percent faster than the sender clock. How many extra bits per second does the receiver receive if the data rate is 1 Mbps?**
(a) 10 bps (b) 100 bps
(c) 1000 bps (d) 10000 bps

Ans. (c) : Receiver clock is 0.1% faster than sender clock. At 1 mbps bits send by sender = 1000000 receiver clock is 0.1% faster than sender clock so extra bit received by receiver = $1000000 \times \frac{0.1}{100}$
= 1000 bits
total bits received by receiver = 1001000 bits
Extra bit per second received by receiver = 1000 bps
Hence option (c) 1000 bps is the correct.

7. **Given U = {1, 2, 3, 4, 5, 6, 7}**
A = {(3, 0.7), (5, 1), (6, 0.8)}
then A will be : (where ~ → complement)
(a) {(4, 0.7), (2, 1), (1, 0.8)}
(b) {(4, 0.3), (5, 0), (6, 0.2)}
(c) {(1, 1), (2, 1), (3, 0.3), (4, 1), (6, 0.2), &, 1)}
(d) {(3, 0.3), (6, 0.2)}

Ans. (c) : Because,
$\mu A'(x) = ((\mu A(x)) = 1 - \mu A(x)$ for all $x \in U$
The given set A will be seen as A = {(1, 0), (2, 0), (3, 0.7), (4, 0), (5, 2) (6, 0.8), (7, 0)}
So the compliment of A will be {(1, 1), (2, 1), (3, 0.3) (4, 2), (6, 0.2) (.7, 2)}
Hence the answer is (c).

8. **Consider a fuzzy set old as defined below**
Old = {(20, 0.1), (30, 0.2), (40, 0.4), (50, 0.6), (60, 0.8), (70, 1), (80, 1)}
Then the alpha-cut for alpha = 0.4 for the set old will be
(a) {(40, 0.4)}
(b) {50, 60, 80}
(c) {(20, 0, 1), (30, 0.2)}
(d) {(20, 0), (30, 0), (40, 1), (50, 1), (60, 1), (70, 1), (80, 1)}

Ans. (d) : Because, alpha cut means it is the cut off value. The number having member ship value more than or equal to in the given fuzzy set will be considered as present in the result. Now in the set OLP person with age 40, 50, 60, 70, 80 are having membership values more than or equal to 0.4 so they are considered as member with value 1 and 2030 with values less than cut off are not member hence they are shown with value 0 so answer is (d).

9. Perception learning, Delta learning and LMS learning are learning methods which falls under the category of

(a) Error correction learning – learning with a teacher
(b) Reinforcement learning – learning with a critic
(c) Hebbian learning
(d) Competitive learning – learning without a teacher

Ans. (a) : Because, Error-correction learning is used with supervised learning, is the technique of comparing the system output to the desired output value and perception learning, delta learning and LMS learning method which falls under this category so the answer is (a).

10. Code blocks allow many algorithms to be implemented with the following parameters :

(a) clarity, elegancy, performance
(b) clarity, elegance, efficiency
(c) elegance, performance, execution
(d) execution, clarity, performance

Ans. (b) : Because, Code blocks allow many algorithm to be implemented with clarity elegance and efficiency. More over, they help the programmer better conceptualize the true nature of algorithm being implement.

11. Match the following with respect to the jump statements :

	List–I		**List–II**
a.	return	i.	The conditional test and increment portions
b.	goto	ii.	A value associated with it
c.	break	iii.	Requires a label for operation
d.	continue	iv.	An exit from only the innermost loop

Codes :

	a	**b**	**c**	**d**
(a)	ii	iii	iv	i
(b)	iii	iv	i	ii
(c)	iv	iii	ii	i
(d)	iv	iii	i	ii

Ans. (a) : Because,

(a) Return : Return the content of a variable after completion of the function body.

(b) go to : Control jumps from one statement to another by following the label.

(c) Break : Stops looping where the break statement appears.

Continue : It increments the loop controlling variable without executing the body of the loop.

⇒ So the correct option is (a).

12. The control string in C++ consists of three important classifications of characters

(a) Escape sequence characters. Format specifies and White-space characters

(b) Special characters, White-space characters and Non-white space characters

(c) Format specifies, White-space characters and Non-white space characters

(d) Special characters, White-space characters and Format specifies

Ans. (c) : Because, In programming language we use unique keywords to store data, just like that, the control string is use to access the different type of keyword, and control strings are the characters that use with the % (percent) and it is a classification of format specifier, white space and non-white space characters.

13. Match the following with respect to I/O classes in object oriented programming :

	List–I		List–II
a.	fopen()	i.	returns end of file
b.	fclose()	ii.	return for any problem report
c.	ferror()	iii.	returns 0
d.	feof()	iv.	returns a file pointer

Codes :

	a	b	c	d
(a)	iv	i	ii	iii
(b)	iii	i	iv	ii
(c)	ii	iii	iv	i
(d)	iv	iii	i	ii

Ans. (a) : Because,

fopen () : Creates a file and open an existing file

Syntax : FILE ∗ fp;

fp = fopen ("<file – name>", "<mode>");

fclose : It closes a file and returns 0 on success and EOF otherwise

Syntax : fclose (<file – pointer>);

ferror ( ) : Checks the error indicator for the given file stream and returning non-zero if set and 0 otherwise

feof ( ) : Checks the end of file indicator for the given file stream returning non-zero if set and 0 otherwise

14. Which one of the following describes the syntax of prolog program?

I. Rules and facts are terminated by full stop(.)

II. Rules and facts are terminated by semicolon (;)
III. Variables names must start with upper case alphabets
IV. Variables names must start with lower case alphabets.

Codes :

(a) I, II (b) III, IV
(c) I, III (d) II, IV

Ans. (c) : Because, in prolog variable begin with an upper case letter. predicate names, function names and the names for objects must begin with a lowercase letter and rules and facts are terminated by full stop (·) So the right answer is option (c).

15. Let L be any language. Define even (W) as the strings obtained by extracting from W the letters in the even-numbered positions and even (L) = {even (W) | W ε L}. We define another language Chop (L) by removing the two leftmost symbols of every string in L given by Chop (L) = $\{W | vW \in L. with |v| = 2\}$. It L is regular language then

(a) even (L) is regular and Chop (L) is not regular.
(b) Both even (L) and Chop (L) are regular.
(c) even (L) is not regular and Chop (L) is regular.
(d) Both even (L) and Chop (L) are not regular.

Ans. (b) : Because, Regular languages are closed under union, intersection, difference, concatenation, kleene closure, Reversal, homomorphism and inverse homomorphism.

A homomorphism on an alphabet is a function that gives a string for each symbol in that alphabet

example h(0) = ab; h(1) = ε

extending of string by $h(a_1 \ldots\ldots. an) = h(a_1) \ldots\ldots h(a_n)$

⇒ So the correct answer is option (b).

16. Software testing is

(a) the process of establishing that errors are not present.
(b) the process of establishing confidence that a program does what it is supposed to do.
(c) the process of executing a program to show that it is working as per specifications.
(d) the process of executing a program with the intent of finding errors

Ans. (d) : Because, Software testing is a process of executing a program or application with the intent of finding errors (the software bugs) and the main objective of software testing is to find any bug present in the software program. Besides, through the process of testing. One can determine the strengths and weaknesses of the program in other criteria such as path testing, dependency testing etc.

17. Assume that a program will experience 200 failures in infinite time. It has now experienced 100 failures. The initial failure intensity was 20 failures/CPU hr. Then the current failure intensity will be

(a) 5 failures/CPU hr.
(b) 10 failures/CPU hr.
(c) 20 failures/CPU hr.
(d) 40 failures/CPU hr.

Ans. (b) : Because, V_0 = 200 failures
μ = 100 failures
λ_0 = 20 failure / CPU hr.
Current failure intensity = $\lambda_0 (1 - \mu/V_0)$
= 20 (1 – 100/200) = 20 (1 – 0.5)
$\Rightarrow$ 10 / failure / CPU hr
and the answer is (b).

18. Consider a project with the following functional units :
Number of user inputs = 50
Number of user outputs = 40
Number of user enquiries = 35
Number of user files = 06
Number of external interfaces = 04
Assuming all complexity adjustment factors and weighing factors as average, the function points for the project will be.

(a) 135 (b) 722
(c) 675 (d) 672

Ans. (d) : Because, External inputs 50 × 4 = 200
External outputs 40 × 5 = 200
External inquiries 35 × 4 = 140
Internal logic files 6 × 10 = 60
External interface files 4 × 7 = 28
Count total = 628
Average value adjustment factor for each Fi = 3
So, $\sum^{14}_{1}$ Fi = 3 × 14 = 42
So, Fp = 628 × (0.65 + 0.01 × 42) = 611.96 ≅ 672
So the answer is (d).

19. Match the following :

	List–I		**List–II**
a.	Correctness	i.	The extent to which a software tolerates the unexpected problems
b.	Accuracy	ii.	The extent to which a software meets its specifications
c.	Robustness	iii.	The extent to which a software has specified functions
d.	Completeness	iv.	Meeting specifications with precision

Codes :

	a	b	c	d
(a)	ii	iv	i	iii
(b)	i	ii	iii	iv
(c)	ii	i	iv	iii
(d)	iv	ii	i	iii

Ans. (a) : Because,

Correctness : Correctness is defined as adherence to the specification that determine how users can interact with the software and how the software should behave when it used correctly. Means, is the software meets its specification.

Accuracy : It is defined as the combination of both types of observational error (random and systematic) So high accuracy requires both high precision and high trueness, means meeting specifications with precision.

Robustness : It is defined as the cope with error during execution and cope with erroneous input means which a software is tolerates the unexpected problems.

Completeness : It is defined as which a software has specified function.

So the answer is (a).

20. Which one of the following is not a definition of error?

(a) It refers to the discrepancy between a computed, observed or measured value and the true., specified or theoretically correct value.

(b) It refers to the actual output of a software and the correct output.

(c) It refers to a condition that causes a system to fail.

(d) It refers to human action that results in software containing a defect or fault.

Ans. (c) : Because, Error mathematical is the difference between the observed value and approximate value and true value of a quantity.

Error is used to refer to human action that results in software containing a defector fault whereas failure refers to a condition that causes a system to fail.

Hence the answer is (c).

21. Which one of the following is not a key process area in CMM level 5?

(a) Defect prevention

(b) Process change management

(c) Software product engineering

(d) Technology change management

Ans. (c) : Because,

CMM (Capability maturity model) : It is framework that describes an evolutionary improvement path for software organizations from an immature process to a mature, discipline one.

The software process is at CMM level 5, if there is continuous process improvement, if there is quantitative feedback from the process and from piloting innovative ideal and technologies. Software processes are the CMM level 5, if there are process change management, detect prevention and technology change management, So option (c) in not a key process area in CMM level 5

22. Consider the following relational schemas for a library database :

Book (Title, Author, Catalog–No, Publisher, Year, Price)

Collection (Title, Author, Catalog–no) with the following functional dependencies :

I. Title, Author → Catalog–no

II. Catalog–no → Title, Author, Publisher, Year

III. Publisher, Title, Year → Price

Assume (Author, Title) is the key for both schemas. Which one of the following is true?

(a) Both Book and Collection are in BCNF.

(b) Both Book and Collection are in 3 NF.

(c) Book is in 2NF and Collection in 3 NF.

(d) Both Book and Collection are in 2 NF.

Ans. (c) : Because, 2NF says that all the non-key attributes should depend the keys, partial dependency (an attribute is defined by a part of a composite key) is not allowed 3NF says that any dependency between non-key attribute should be cleared.

BCNF says that all determinants in a table is key for every non trivial (no overlapping attributes) function. In the table book, (Title Author) identifies catalog-no. Which if turn identifies publisher and year further (publisher, Title, Year) identifies price. Now here the dependents are partially dependent on the key attributes on left hand side of (R). So the table book is in 2NF. In the table collection, (Title, Author) determines catalog-no., the only non-key attribute. So it is in 3NF as well as BCNF.

23. Specialization Lattice stands for

(a) An entity type can participate as a subclass in only one specialization.

(b) An entity type can participate as a subclass in more than one specialization.

(c) An entity type that can participate in one specialization.

(d) An entity type that can participate in one generalization.

Ans. (b) : Because,

Lattice : It is a partially-ordered set (POSET) that has a unique SUPREMUM and a unique INFIMUM for all pairs of elements.

Specialization : It is a process of creating one or more subclasses of an entity type that should have some attributes uncommon with the given super-class. In specialization lattice, one type of subclass can have one or more types of super class whereas in Hierarchal lattice, one type of subclass can have only one type of super class.

24. Match the following :

	List–I		List–II
a.	Timeout ordering protocol	i.	Wait for graph
b.	Deadlock prevention	ii.	Roll back
c.	Deadlock detection	iii.	Wait-die scheme
d.	Deadlock recovery	iv.	Thomas Write Rule

Codes :

	a	b	c	d
(a)	iv	iii	i	ii
(b)	iii	ii	iv	i
(c)	ii	i	iv	iii
(d)	iii	i	iv	iii

Ans. (a) : Because,
(a) Time Ordering Protocol : If an old transaction Ti has time stamp Ts(Ti), a new transaction Tj is assigned time stamp. It is based Thomas write rule.
(b) Deadlock Prevention : It is based on wait-die scheme means if Ts(Ti) < Ts(Tj), Ti is allowed to wait, otherwise, if Ts(Tj) < Ts(Ti) is aborted later started.
(c) Deadlock Detection : It can be done with the help of wait-for graph that corresponds to local or non local processes that either hold or request local resources. A wait for graph resource allocation state.
(d) Deadlock Recovery : Deadlock can be recovered with a roll back of a transaction that will eventually release a resource from a process.
Hence the answer is (a).

25. Consider the schema
$R = \{S, T, U, V]$
and the dependencies
$S \rightarrow T, T \rightarrow U, U \rightarrow V$ and $V \rightarrow S$
If $R = (R_1$ and $R_2)$ be a decomposition such that $R_1 \cap R_2 = \phi$ then the decomposition is
(a) not in 2 NF
(b) in 2 NF but not in 3 NF
(c) in 3 NF but not in 2 NF
(d) in both 2 NF and 3 NF

Ans. (d) : Because, $R_1 \cap R_2 \neq \phi$. This makes the decomposition lossless join, as all the attributes are keys, $R_1 \cap R_2$ will be a key of the decomposed relation Now, even the original relation R is in 3NF (Even BCNF) as all the attributes are prime attributes (in fact attribute is a candidate key) Hence, any decomposition will also be in 3NF and 2NF, So the answer is option (d).

26. Which one of the following is not a Client-Server application?
(a) Internet chat (b) Web browser
(c) E-mail (d) Ping

Ans. (D) : Ping is not a client server application. Ping is a computer network administration unility used to test the reachability of a host on an Internet protocol ping there is no server that provides a service.

27. **Which of the following concurrency protocol ensures both conflict serializability and freedom from deadlock :**

I. 2-phase locking

II. Time phase ordering

(a) Both I & II (b) II only

(c) I only (d) Neither I nor II

Ans. (b) : Because, 2 phase locking is a concurrency control method that guarantees serializability. The protocol utilizes locks, applied by a transaction to a data, which may block other transaction from accessing the same data during the transactions life. 2PL may be lead to deadlocks that results from the mutual blocking of two or more transaction.

Timestamp-based concurrency control is a non lock concurrency method. In timestamp based method, deadlock occur as no transaction ever waits.

28. **Match the following :**

	List–I		**List–II**
a.	Expert systems	i.	Pragmatics
b.	Planning	ii.	Resolution
c.	Prolog	iii.	Means-end analysis
d.	Natural language processing	iv.	Explanation facility

Codes :

	a	**b**	**c**	**d**
(a)	iii	iv	i	ii
(b)	iii	iv	ii	i
(c)	i	ii	iii	iv
(d)	iv	iii	ii	i

Ans. (d) : Because,

(a) Expert System : Most of the expert system have a module called explanation facility that helps the system to allow the user to ask a question and how the module reached the goal.

(b) Planning : Means-end-analysis is useful for planning activities.

(c) Prolog : Resolution is a mechanism for interference from two or more clauses.

(d) Natural Language Processing : Pragmatic analysis means re-interpreting the sentence to find out the overall communicative and social context of the input text. So the answer is (d).

29. **STRIPS addresses the problem of efficiently representing and implementation of a planner. It is not related to which one of the following?**

(a) SHAKEY (b) SRI

(c) NLP (d) None of these

Ans. (c) : STRIPs was devised **SRI** in the early 1970s to control a robot called **SHAKEY** strips is not related with NLP (Natural Language Processing).

30. **Slots and facets are used in**

(a) Semantic Networks (b) Frames

(c) Rules (d) All of these

Ans. (b) : Because,
Frames : Frames are structured objects that represent knowledge and contain labeled slots with their values.
Slots : Slots are almost similar to attributes as in ERD or object-oriented systems in the sense that they may contain declarative as well as procedural information.
Facets : Facets are almost similar to description to the slot values that may function as criteria or validity of the values inserted into a slop.

Name of frame		
Slots	Slot valves	Facets

31. Consider f (N) = g(N) + h(N)
Where function g is a measure of the cost of getting from the start node to the current node N and h is an estimate of additional cost of getting from the current node N to the goal node. Then f(N) = h(N) is used in which one of the following algorithms?
(a) A* algorithm
(b) AO* algorithm
(c) Greedy best first search algorithm
(d) Iterative A* algorithm

Ans. (c) : Because, the greedy algorithm work in a way that it has same estimate (called a heuristic) of how for from the goal any vertex is. Instead of selecting the vertex closet to the starting point, it selects the vertex closet to the goal, so the answer is (c).

32. ———— predicate calculus allows quantified variables to refer to objects in the domain of discourse and not to predicates or functions.
(a) Zero-order (b) First-order
(c) Second-order (d) High-order

Ans. (b) : Because, first-order logic is a collection of formal systems used in mathematics, philosophy and computer science. It also known as first-order predicate calculus the lower predicate calculus, quantification theory & the predicate logic first order logic uses quantified variable over objects. It allows the use of sentence that contain variables.

33. ———— is used in game trees to reduce the number of branches of the search tree to be traversed without affecting the solution.
(a) Best first search
(b) Goal stack planning
(c) Alpha-beta pruning procedure
(d) Min-max search

Ans. (c) : Because, It is a search algorithm that seeks to decrease the number nodes that are evaluated by the Min-max algorithm in its search free. It is an adversial search algorithm used commonly for machine playing of two-player games.

34. **Consider a uniprocessor system where new processes arrive at an average of five processes per minute and each process needs an average of 6 seconds of service time. What will be the CPU utilization?**

(a) 80% (b) 50%

(c) 60% (d) 30%

Ans. (b) : Because, Given in system there are on average 5 new process arrive pertime.

So, arrival rate = 5 process / min

In every 12 second a $\left(\frac{60}{5}\right)$ new process arrive is system or we can say every process stays 12 second with the CPU.

Give service time = 6 seconds

$$\text{CPU utilization} = \frac{\text{Service time}}{\text{Staring time}} \times 100$$

$$= \frac{6}{12} \times 100 = 50\%$$

Hence the right option is (b).

35. **Consider a program that consists of 8 pages (from 0 to 7) and we have 4 page frames in the physical memory for the pages. The page reference string is :**

1 2 3 2 5 6 3 4 6 3 7 3 1 5 3 6 3 4 2 4 3 4 5 1

The number of page faults in LRU and optimal page replacement algorithms are respectively (without including initial page faults to fill available page frames with pages) :

(a) 9 and 6 (b) 10 and 7

(c) 9 and 7 (d) 10 and 6

Ans. (b) : Because, first we finds the LRU page replacement

1	2	3	2	5	6	3	4	6	3	7	3	1	5	3	6	3	4	2	4	3

then,

				5	5	5	5	5	5	7	7	7	7	7	6	6	6	6	6	6
		3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	2	3	3
	2	2	2	2	2	2	4	4	4	4	4	1	1	1	1	1	4	4	4	4
1	1	1	1	1	6	6	6	6	6	6	6	6	5	5	5	5	5	2	2	2
					F		F			F		F	F		F		F	F		

So page faults

$\Rightarrow$ 10, means total replacement at LRU is 10.

Now, we final the optimal page replacement of given above question.

				5	5	5	4	4	4	7	7	7	5	5	5	5	5	5	5
		3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
	2	2	2	2	6	6	6	6	6	6	6	6	6	6	6	4	4	4	4
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	2	2	2
					F		F			F			F			F			

So total replacement at optimal page replacement is 7.

So the correct answer is (b).

36. **Which of the following statements is not true about disk-arm scheduling algorithms?**

(a) SSTF (shortest seek time first) algorithm increases performance of FCFS.
(b) The number of requests for disk service are not influenced by file allocation method.
(c) Caching the directories and index blocks in main memory can also help in reducing disk arm movements.
(d) SCAN and C-SCAN algorithms are less likely to have a starvation problem.

Ans. (b) : Because, option (b) is NOT CORRECT as the number of request for disk drive can be greatly influenced by file allocation method So the answer is (b).

37. ——— maintains the list of free disk blocks in the Unix file system

(a) I-node (b) Boot block
(c) Super block (d) File allocation table

Ans. (c) : Because, Super block describes the state of file system, the total size of partition, the block size, pointers to a list of free blocks, the inode number of root directory, magic number etc, so the answer is (c).

38. A part Windows 2000 operating system that is not portable is

(a) Device Management
(b) Virtual Memory Management
(c) Processor Management
(d) User Interface

Ans. (b) : Because, The design of virtual memory assumes that the underlying hardware supports virtual to physical mapping.
So the answer is (b).

39. Match the following with reference to Unix shell scripts :

	List–I		**List–II**
a.	$?	i.	File name of the current script
b.	$#	ii.	List of arguments
c.	$0	iii.	The number of arguments
d.	$*	iv.	Exit status of last command

Codes :

	a	**b**	**c**	**d**
(a)	iii	ii	i	iv
(b)	ii	iii	i	iv
(c)	iv	iii	i	ii
(d)	i	iii	i	iv

Ans. (c) : Because,
$? : This command shows exit status of the last command executed.
$# : This command shows number of argument supplied to a script.
$O : This command shows filename of current script.
$* : This command shows all the arguments are double quoted.

40. The advantage of ——— is that it can reference memory without paying the price of having a full memory address in the instruction.

(a) Direct addressing
(b) Indexed addressing
(c) Register addressing
(d) Register Indirect addressing

Ans. (d) : Because, Register indirect addressing means that the location of an operand is held in a register. It is also called indexed addressing Register indirect addressing modes requires three read operation to access an operand. Hence the answer is (d).

41. The reverse polish notation equivalent to the infix expression
((A + B) * C + D)/(E + F + G)

(a) A B + C * D + EF + G + /
(b) A B + C D * + E F + G + /
(c) A B + C * D + E F G + +/
(d) A B + C * D + E + F G +/

Ans. (a) : Because, ((A + B) ∗ C + D) / (E + F + G)
(AB + ∗C + D) / (E + F + G)
(AB + C ∗ + D) / (E + F + G)
AB + C ∗ D + / (E + F + G)
AB + C ∗ D) + / (EF + G)
AB + C ∗ D + / EF + G +
AB + C ∗ D + EF + G + /
Hence the answer is (a).

42. The output of a sequential circuit depends on

(a) present input only
(b) past input only
(c) both present and past input
(d) past output only

Ans. (c) : Because, There are 2 types of circuit
(1) Combinational, (2) Sequential
Combinational circuit are those whose outputs depends upon correct input like ADDER, SUBTRACER, MUX, DEMUX etc.
Sequential circuits are those whose output depends on both current and previous input.
Hence the answer is (c).

43. A byte addressable computer has a memory capacity of 2^m K bytes and can perform 2^n operations. An instruction involving 3 operands and one operator needs a maximum of

(a) 3m bits
(b) m + n bits
(c) 3m + n bits
(d) 3m + n + 30 bits

Ans. (d) : Because, Memory capacity is 2^m Kbytes = 2^{m+10} bytes. Whole memory can be represented by M + 10 bits.2^n operations can be represented by n bits.
Now instruction of from
Instruction OP_1, OP_2, OP_3 will take (3m + 30 + n) bits.
So the answer is (d).

44. **Which of the following flip-flops is free from race condition?**
(a) T flip-flop
(b) SR flip-flop
(c) Master-slave JK flip-flop
(d) None of the above

Ans. (c) : Because, In the case of JK FF when
J = 1 K = 1 the Q toggle to 0 and 1.
It happens continuously fill the clock is enable ...... and prediction of Q is uncertain (race condition)...... Master-slave JK flip flop is used to resolve this problem.

45. **One of the main features that distinguish microprocessor from micro-computers is**
(a) words are usually larger in microprocessors.
(b) words are shorter in microprocessors.
(c) microprocessor does not contain I/O devices.
(d) None of the above.

Ans. (c) :A microprocessor is a computer processor which incorporates the functions of a computer's control processing unit (CPU) on a single integrated circuit (IC), or at most a few integrated circuits.
A microcomputer is small, relatively, inexpensive computer with a microprocessor as its control processing unit (CPU). It includes a microprocessor, memory, and minimal input/output (I/O) circuitry mounted on a single printed circuit board.
That means, microprocessor does not contain input/output devices.

46. **The output generated by the LINUX command :**
$ seq 1 2 10
will be
(a) 1 2 10
(b) 1 2 3 4 5 6 7 8 9 10
(c) 1 3 5 7 9
(d) 1 5 10

Ans. (c) : In linux seq command is used to print sequence of number seq command print numbers from first to last, in steps of increment.
If first or incremented is omitted, it defaults to 1. That is, an emitted INCREMENT defaults to 1 even when last is smaller than FIRST. FIRST, INCREMENT, and LAST are interpreted as floating point values INCREMENTED is usually positive if FIRST is smaller than LAST, and INCREMENTED is usually negative if first is greater than LAST.
then,

$ seq	First	increment	last
	↑	↑	↑
	1	2	10

then the output sequence generated is = 13579
⇒ Hence the correct choice is (c).

47. **All the classes necessary for windows programming are available in the module :**
(a) win.txt
(b) win.main
(c) win.std
(d) MFC

Ans. (a) : Because, win·txt is a group of classes that is use for windows programming.

48. Windows 32 API supports

(a) 16-bit Windows (b) 32-bit Windows

(c) 64-bit Windows (d) All of the above

Ans. (d) : Because, API (Application Program Interface) Index are formally called the win32 API. The name windows API more accurately reflects its root in 16-bit windows and in support on 64-bit windows. Hence the correct choice is (d).

49. Superficially the term "object-oriented", means that, we organize software as a

(a) collection of continuous objects that incorporates both data structure and behaviour.

(b) collection of discrete objects that incorporates both discrete structure and behaviour.

(c) collection of discrete objects that incorporates both data structure and behaviour.

(d) collection of objects that incorporates both discrete data structure and behaviour.

Ans. (c) : Because, object oriented development makes software closet to abstractions the exists in the real world and the term object oriented means that software is a collection of discrete objects that incorporate both data structure and behavior.

50. The "part-whole", or "a-part-of", relationship in which objects representing the components of something associated with an object representing the entire assembly is called as

(a) Association (b) Aggregation

(c) Encapsulation (d) Generalisation

Ans. (b) : Because, Aggregation is defined as a group or mass of distinct or varied things, person, etc and closets related concept of composition is said to aggregation. Hence the answer is (b).

51. The pure object oriented programming language with extensive metadata available and modifiable at run time is

(a) Small talk (b) C++

(c) Java (d) Eiffel

Ans. (a) : Because, Small talk was one of the earliest OOP's language (with other like simulate and Eiffel) and can be said to be extremely "Pure" in an OOP sense.

In small talk everything is an object and objects are only communicated with via the sending of message. Hence the answer is (a).

52. Match the following interfaces of Java. Servlet package :

	List–I		**List–II**
a.	Servlet Config	i.	Enables Servlets to log events
b.	Servlet Context	ii.	Read data from a client
c.	Servlet Request	iii.	Write data to a client
d.	Servlet Response	iv.	To get initialization parameters

Codes :

	a	b	c	d
(a)	iii	iv	ii	i
(b)	iii	ii	iv	i
(c)	ii	iii	iv	i
(d)	iv	i	ii	iii

Ans. (d) : Because,

(a) Servlet Config : Servlet config is created by the web container for each servlet and this object can be used to get configuration information and get initialization parameters.

(b) Servlet Context : It is created by the web container at time of developing the project and this can be used to get configuration information from web. XML file and it enables servlets to log events.

(c) Servlet Request : This is used to provide the client request information to a servlet such as content type, content length etc mean it used for read data from a client.

(d) Servlet Response : Servlet response is used to write data to a client.

⇒ Hence the answer is (d).

53. The syntax of capturing events method for document object is

(a) CaptureEvents()

(b) CaptureEvents(Orgs eventType)

(c) CaptureEvents(eventType)

(d) CaptureEvents(eventVal)

Ans. (c) : Capture events (Event type) method instructs the window to capture all events of a particular type.

54. Linking to another place in the same or another webpage require two A (Anchor) tags, the first with the ________ attribute and the second with the _______ attribute.

(a) NAME & LINK

(b) LINK & HREF

(c) HREF & NAME

(d) TARGET & VALUE

Ans. (c) : Because, Name attribute specifies the name of an anchor and it is used to create a bookmark inside a document.

Syntax : <a name = "value">

⇒ href attribute specifies the URL of the page the link goes to, and if the href attribute is not present, the <a> tag is nat a hyperlink.

Syntax : <a href = "URL">

55. Given an image of size 1024 × 1024 pixels in which intensity of each pixel is an 8-bit quality. It requires _______ of storage space if the image is not compressed.

(a) one Terabyte (b) one Megabyte

(c) 8 Megabytes (d) 8 Terabytes

Ans. (b) : Because, Each pixel needs to store 1 Byte (8 bit)

Total 1024 × 1024 pixels

Then,

Total storage space = $1024 \times 1024 \times 1 = 2^{10} \times 2^{10} \times 1 = 2^{20}$ = 1 MB

So the answer is (b).

56. Match the following cryptographic algorithms with their design issues :

List–I		List–II	
a.	DES	i.	Message Digest
b.	AES	ii.	Public Key
c.	RSA	iii.	56-bit key
d.	SHA-1	iv.	128-bit key

Codes :

	a	b	c	d
(a)	ii	i	iv	iii
(b)	iii	i	iv	ii
(c)	iii	iv	ii	i
(d)	iv	i	ii	iii

Ans. (c) : Because,

(a) DES : DES is an implementation of a fiestel cipher. It uses 16 round fiestel structure. DES has an effective key length of 56 bits.

(b) AES : The more popular and widely adopted symmetric encryption algorithm is AES. The features of AES is symmetric key symmetric block cipher and it has 128 bit key.

(c) RSA : RSA algorithm works on two different keys i.e public key & private key. As the name public key given to everyone & private key is kept private.

(d) SHA-1 : It is message digest.

57. Consider a code with five valid code words of length ten :
0000000000, 0000011111,
1111100000, 1110000011,
1111111111
Hamming distance of the code is

(a) 5 (b) 10
(c) 8 (d) 9

Ans. (a) : Because, Max hamming distance = 10(between 1st and 5th 82nd 83rd)

min, hamming distance = 4(between 384)

1111100000
1110000011

So the answer is (a).

58. Which of the following special cases does not require reformulation of the problem in order to obtain a solution?

(a) Alternate optimality
(b) Infeasibility
(c) Unboundedness
(d) All of the above

Ans. (a) : Because, in alternative optimal solution is also called an alternative optima, which is when a linear programming problem has more than one optimal solution an optimal solution is a solution to a problem. Which satisfies the set of constraints of the problem and the objective function which is to maximize and minimize. So the answer is (a).

59. The given maximization assignment problem can be converted into a minimization problem by

(a) subtracting each entry in a column from the maximum value in that column.
(b) subtracting each entry in the table from the maximum value in that table.
(c) adding each entry in a column from the maximum value in that column.
(d) adding maximum value of the table to each entry in the table.

Ans. (b) : Because, There are problem where certain facilates have to be assigned to a number of jobs so as to maximize the overall performance of the assignment. The problem can be converted into a minimization problem in the following wars and then Hungarian method can be used for its solution

(1) Change the signs of all values given in table.

(2) Select the highest element in the entire assignment table and subtract all the elements of the table from the highest element.

⇒ So the answer is (b).

60. The initial basic feasible solution of the following transportion problem :

Destination

		D_1	D_2	D_3	Supply
	O_1	2	7	4	5
Origins	O_2	3	3	1	8
	O_3	5	4	7	7
	O_4	1	6	2	14
Demand		7	9	18	

is given as

5		
		8
	7	
2	2	10

then the minimum cost is

(a) 76 (b) 78
(c) 80 (d) 82

Ans. (a) : Because,

Cost Matrix v_i

2	7	4	$v_1 = 1$
3	3	1	$v_2 = -1$
5	4	7	$v_3 = -2$
1	6	2	$v_4 = 0$

v_j $v_1 = 1$, $v_2 = 6$ $v_3 = 2$

unallocated cell penalties ($P_{ij} = v_i + v_j - c_{ij}$)

$P_{12} = 1 + 6 - 7 = 0$

$P_{13} = 1 + 2 - 4 = -1$

$P_{21} = -1 + 1 - 3 = -3$

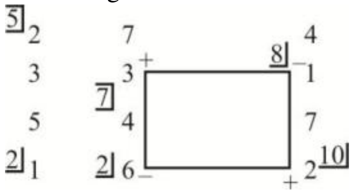
$P_{22} = -1 + 6 - 3 = ②$

$P_{31} = -2 + 1 - 5 = -6$

$P_{33} = -2 + 2 - 7 = -7$

$\therefore$ So P_{22} is identified ccr

Thus P_{22} should now be allocated which would result in cost minimizing



$v_1 = 0$ $v_2 = 3$ $v_3 = 1$

5]2	+2]7	4	$v_1 = 2$
3	1]3	$\underline{8-2=6}]1$	$v_2 = 0$
5	4	7	$v_3 = 1$
2]1	$\underline{2-2=0}]6$	$\underline{10+2=12}]2$	$v_4 = 1$

Now penalties for non allocated cells are

$$\left.\begin{aligned} P_{12} &= 2+3-7=-2 \\ P_{13} &= 2+1-4=-1 \\ P_{21} &= 0+0-3=-3 \\ P_{31} &= 1+0-5=-4 \\ P_{33} &= 1+1-7=-5 \\ P_{42} &= 1+3-6=-2 \end{aligned}\right\} \text{Optimel solu.}$$

Now cost are,

$5 \times 2 + 3 \times 2 + 1 \times 6 + 4 \times 7 + 2 \times 12 + 1 \times 2$

$= 1 + 6 + 6 + 28 + 24 + 2$

$= 76$

So the answer is (a).

61. Given the following equalities :

$E_1 : n^{K+\epsilon} + n^K \lg n = \theta(n^{K+\epsilon})$ for all fixed K and ϵ. $K \geq 0$ and $\epsilon > 0$.

$E_2 : n^3 2^n + 6n^2 3^n = O(n^3 2^n)$

Which of the following is true?

(a) E_1 is correct and E_2 is correct.

(b) E_1 is correct and E_2 is not correct.

(c) E_1 is not correct and E_2 is correct.

(d) E_1 is not correct and E_2 is not correct.

Ans. (b) : Because, for E_2;

find which one is large

N ^ 3 * 2 ^ N = 6N ^ 2 * 3 ^ N

N * 2 ^ N = 6 * 3 ^ N (Now take log on both side)

Log(N * 2 ^ N) = Log(6 * 3 ^ N)

Log N + N Log 2 = Log 6 + N Log 3

Log N + N = 1.5N (Log6 = Constant m Log3 = 1.5)

Log N = 0.5 N

O(0.5N) = O(6 * N ^ 2 * 3 ^ N)

So E_2 is false.
for E_1,
Compare N ^ E = log n
Since E > 0 N ^ E is large compared to log n
So E_1 is correct.
Since the right option is (b).

62. Consider the fractional knapsack instance n = 4, (p_1, p_2, p_3, p_4) = (10, 10, 12, 18), (w_1, w_2, w_3, w_4) = (2, 4, 6, 9) and M = 15. The maximum profit is given by
(Assume p and w denotes profit and weight of objects respectively)
(a) 40 (b) 38
(c) 32 (d) 30

Ans. (b) : Because,

P_1 / W_1	5
P_2 / W_2	2.5
P_3 / W_3	2
P_4 / W_4	2

Now select the one which has max (P/W) ratio that is P_1/W_1 = 5 So select 10
Next P_2/W_2 = 2.5 select 10
Now P_3/W_3 and P_4/W_4 has same ratio but P_4 gives maximum profit 80 select P_4,
therefore total wcign = (2 + 4 + 9) = 15
8 Max profit = 10 + 10 + 18 = 38
So the answer is (b).

63. The solution of the recurrence relation of T(n) $= 3T\left(\text{floor}\left(\frac{n}{4}\right)\right) + n$ is
(a) $O(n^2)$ (b) O(n/g n)
(c) O(n) (d) O(/ g n)

Ans. (c) : Given recurrence relation is:

$$T(n) = 3T\left(\left\lceil \frac{n}{4} \right\rceil\right) + n$$

Using master theorem case 3 is:
$(n^{\log b^{(a)}}) \leq f(n)$ then $T(n) = \Theta(n)$
Given a = 3, b = 4 and f(n) = n then
$(n^{\log 4^{(3)}}) \leq n$, then $T(n) = \Theta(n)$
So, $T(n) = O(n) = O(n^2)$
Both option (a) and (c) are true, but option (c) is best choice.

64. If h is chosen from a universal collection of hash functions and is used to hash n keys into a table of size m, where n ≤ m, the expected number of collisions involving a particular key K is
(a) less than 1 (b) less than *l*g n
(c) greater than 1 (d) greater than *l*g n

Ans. (a) : Because, This is a theorem in universal hashing :
"if n is chosen from a universal collection and is used for hash n keys in table of size m, where n < = m, the expected number of collision involving a particular key is less than 1")
So the answer is (a).

65. Given the following statements :

S_1 : The subgraph-isomorphism problem takes two graphs G_1 and G_2 and asks whether G_1 is a subgraph of G_2.

S_2 : The set-partition problem takes as input a set S of numbers and asks whether the numbers can be partitioned into two sets A and $\bar{A} = S - A$ such that

$$\sum_{x \in A} x = \sum_{x \in \bar{A}} x$$

Which of the following is true ?

(a) S_1 is NP problem and S_2 is P problem.
(b) S_1 is NP problem and S_2 is NP problem.
(c) $S_1$is P problem and S_2 is P problem.
(d) S_1 is P problem and S_2 is NP problem.

Ans. (b) : Because,
⇒ S_2 is a MP problem means it is solved in a polynomial time.
⇒ S_1 is a P problem means if exists at least one algorithm to solve that problem.
⇒ So the answer is (b).

66. Suppose that the splits at every level of quicksort are in the proportion (1 − α) to α. where 0 < α ≤ 1/2 is a constant. The minimum depth of a leaf in the recursion tree is approximately given by

(a) $-\dfrac{lg n}{lg(1-\alpha)}$ (b) $-\dfrac{lg(1-\alpha)}{lg n}$

(c) $-\dfrac{lg n}{lg \alpha}$ (d) $-\dfrac{lg \alpha}{lg n}$

Ans. (a) : Because, if we split with α will get minimum depth. Suppose α = 1/2 means pivot is the middle element, with such case recursion tree will be with minimum depth.
(1) na ^ k = 1
k = logn / logα
(2) n(1 − α) ^ k = 1
k = logn / log(1 − α)
So the correct answer is (a).

67. Ten signals, each requiring 3000 Hz, are multiplexed on to a single channel using FDM. How much minimum bandwidth is required for the multiplexed channel? Assume that the guard bands are 300 Hz wide.

(a) 30,000 (b) 32,700
(c) 33,000 (d) None of the above

Ans. (b) : Because, Ten signal need $10 \times 3000 = 30000$ Hz

9(10 − 1) Guard bands (or gaps) needs = $300 \times 9 = 2700$ Kz

So minimum bandwidth need 30000 + 2700 = 32700 Hz

So the answer is (b).

68. A terminal multiplexer has six 1200 bps terminals and 'n' 300 bps terminals connected to it. If the outgoing line is 9600 bps, what is the value of n?

(a) 4 (b) 8

(c) 16 (d) 28

Ans. (b) : Because, the outgoing line = 9600 pbs terminal multiplexer has six 1200 pbs so total traffic $(1200 * 6) = 7200$ bps and then are n 300 bps terminals

So $7200 + 300 * n = 9600 \Rightarrow 300n = 2400 \Rightarrow n = 8$

So the answer is (b).

69. Which of the following is used in the options field of IPv4?

(a) Strict source routing

(b) Loose source routing

(c) time stamp

(d) All of the above

Ans. (d) : Because, IPv4 is the fourth version in the development of the IP. and IPv4 use loose source for an IP option can be used for address translation and IPv4 also use strict source routing for the contrast with loose source routing, in every step of the route is decided in advance where the packet is send and IPv4 also time stamp.

So the correct answer is (d).

70. Which layers of the OSI reference model are host-to-host layers ?

(a) Transport, Session, Presentation, Application

(b) Network, Transport, Session, Presentation

(c) Data-link, Network, Transport, Session

(d) Physical, Data-link, Network, Transport

Ans. (a) : Because, The host-to-host layers main purpose is to shield the upper layer application from the complexity of network.

	Layer	Protocol data unit (PDU)
40St layers	7. Application	Data
	6. Presentation	
	5. Session	
	4. Transport	Segment (TCP) Datagram (UDP)
Media layers	3. Network	Packet
	2. Data link	Frame
	1. Physical	Bit

So the answer is (a).

71. A network on the internet has a subnet mask of 255, 255, 240, 0. What is the maximum number of hosts it can handle?

(a) 1024 (b) 2048

(c) 4096 (d) 8192

Ans. (none of these) :
Because, **Subnet mask :-** 255.255.240.0

$\underbrace{11111111 \cdot 11111111 \cdot 1111}_{\text{Act ID}}\underbrace{0000 \cdot 00000000}_{\text{Host ID}}$

It is a class B-Network for a class B

Network, the upper 16 bits form the network address and lower 16 bit are subnet and host fields of the lower 16 bits most significant bit are 1111. This haves 12 bits for the host number, so $4096(2^{12})$ host address exists First and last address are special so the maximum number of address.

$\Rightarrow 4096 - 2 = 4094$, So the answer is (none of these).

72. Four bits are used for packed sequence numbering in a sliding window protocol used in a computer network. What is the maximum window size?

(a) 4 (b) 8

(c) 15 (d) 16

Ans. (c) : Because, In case of Go Back N protocol sender window size is n – 1 and receiver window size is 1 the size is (2 ^ n) – 1

So the answer is 15.

73. Given the following two grammars :

$\mathbf{G_1 : S \rightarrow AB \mid aaB}$

$\mathbf{A \rightarrow a \mid Aa}$

$\mathbf{B \rightarrow b}$

$\mathbf{G_2 : S \rightarrow a\,S\,b\,S \mid b\,S\,a\,S \mid \lambda}$

Which statement is correct ?

(a) G_1 is unambiguous and G_2 is unambiguous.

(b) G_1 is unambiguous and G_2 is ambiguous.

(c) G_1 is ambiguous and G_2 is unambiguous.

(d) G_1 is ambiguous and G_2 is ambiguous.

Ans. (d) : Because, $\mathbf{G_1}$ **:** $3 \rightarrow AB \mid aab$

$A \rightarrow a \mid Aa$

$B \rightarrow b$

To generate string "aab"

$S \rightarrow AB \rightarrow AaB \rightarrow aaB \rightarrow aab$

$S \rightarrow aaB \rightarrow aab$

So ambiguous

$\mathbf{G_2}$ **:** $S_2 \rightarrow aSbS \mid bSAs \mid \lambda$

generate "abab"

$S \rightarrow aSbS \rightarrow abS \rightarrow abaSbS \rightarrow abab$

$S \rightarrow aSbS \rightarrow aSb \rightarrow abSaSb \rightarrow abab$

So ambiguous

$\Rightarrow$ So the correct answer is (d).

74. Match the following :

	List–I		List–II
a.	Chomsky Normal form	i.	$S \rightarrow b\ S\ S \mid a\ S \mid c$
b.	Greibach Normal form	ii.	$S \rightarrow a\ S\ b \mid ab$
c.	S-grammar	iii.	$S \rightarrow AS \mid a$ $A \rightarrow SA \mid b$
d.	LL grammar	iv.	$S \rightarrow a\ B\ S\ B$ $B \rightarrow b$

Codes :

	a	b	c	d
(a)	iv	iii	i	ii
(b)	iv	iii	ii	i
(c)	iii	iv	i	ii
(d)	iii	iv	ii	i

Ans. (c) : Because,

(a) Chomsky Normal Form : Its production rules should be of the form

$A \rightarrow BC$, or

$A \rightarrow a$

$S \rightarrow \varepsilon$

iii satisfies this

(b) Greiback Normal Form : Its production rules should be of form

$A \rightarrow aA_1A_2$ — An or

$A \rightarrow a$

iv satisfies this

(c) S-Grammar : If every production's right hand side begins with a terminal signal and this terminal is different for any two productions with the left hand side i satisfies this

(d) LL Grammar : It two production rules can begin with same terminal symbols if they have same select sets.

So ii satisfies this then the right answer is (c).

75. Given the following two languages :

$L_1 = \{a^n b^n \mid n \geq 1\} \cup \{a\}$

$L_2 = \{w\ C\ w^R \mid w \in \{a, b\}\}$

Which statement is correct ?

(a) Both L_1 and L_2 are not deterministic.

(b) L_1 is not deterministic and L_2 is deterministic.

(c) L_1 is deterministic and L_2 is not deterministic.

(d) Both L_1 and L_2 are deterministic.

Ans. (d) : Because, L_2 ...... insert symbol into stack until C doesn't appear...... if C the skip it and compare top of stack with input symbol .... Because of C it becomes deterministic L_1 is also deterministic because, if "a" comes then we can't divide .... but after "a", "a" or "b" comes then first part of given language is considered.

So the correct answer is (d).